

THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MSBA7012 / MACC7022 / PMAA7017 Social Media and Digital Marketing Analytics

GENERAL INFORMATION
Instructor: Prof. Hailiang Chen Email: chen19@hku.hk Office: KKL 840 Phone: 3917 0016 Consultation Hours: TBA Teaching Assistants: TBA Pre-requisites: None Course Website: Moodle Software tools used in this course: • AI-assisted coding: GitHub Copilot and Azure OpenAI Assistants with Code Interpreter • Python programming: PyCharm Community (a Python integrated development environment or IDE) or VS Code
COURSE DESCRIPTION
Social media has become an important data source for understanding the behaviors of business stakeholders, such as customers, suppliers, partners, investors, and employees. Analyzing social media data can reveal useful insights for strategic planning and help achieve various business objectives. This course is designed to help students develop strategies to create and extract value from applying social media analytics. Techniques covered will include web scraping, data management, machine learning, social network analytics, text analytics, natural language processing, sentiment analysis, and knowledge mining. Students will also learn how to carry out a business analytics project from beginning to end.
COURSE OBJECTIVES
1. To provide an understanding of key fundamentals & principles of social media and digital marketing analytics 2. To provide an understanding of the role of social media and digital marketing strategies in achieving business objectives 3. To develop expertise in designing & deploying social media and digital marketing strategies 4. To apply social media analytics to address business needs through hands-on practical applications
Programme Learning Outcomes
PLO1: Acquisition and internalization of knowledge of the programme discipline PLO2: Application and integration of knowledge PLO3: Inculcating professionalism and leadership in the interdisciplinary fields of accounting analytics PLO4: Developing global outlook PLO5: Mastering communication skills
COURSE LEARNING OUTCOMES

Course Learning Outcomes		Aligned Programme Learning Outcomes	
CLO1 – Learn strategic & technical aspects of social media and digital marketing analytics		PLO1, 2, 3, 4	
CLO2 – Demonstrate understanding of metrics to assess effectiveness of digital marketing & social media campaigns		PLO1, 2	
CLO3 – Collect, analyze, derive and present insights from social media and digital marketing data		PLO1, 2, 3, 4, 5	
CLO4 – Demonstrate real world application of tools & techniques related to social media and digital marketing analytics		PLO1, 2, 3, 4, 5	
CLO5 - Integrate and apply key concepts covered under four prior learning outcomes		PLO1, 2, 3, 4, 5	
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures and presentations		33	27.5%
T&L2. Online participation, including Q&A, helping others solve problems		6	5%
T&L3. Assignments, tutorials, and group project		51	42.5%
T&L4. Lecture preparation & self-study		30	25%
Total		120	100%
Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. In-class and Online Participation		20%	CLO1, 2, 3, 4, 5
A2. Group Project		50%	CLO1, 2, 3, 4, 5
A3. Individual Assignments		30%	CLO1, 2, 3, 4, 5
	Total	100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments		
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good performance in assignments		
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable performance in assignments		

D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable performance in assignments
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments and the Group Project	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills

D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

The course will cover the following topics. A detailed teaching schedule will be provided in due course.

- Introduction & Code Interpreter Basics
 - Azure OpenAI Assistants with Code Interpreter
- Predicting Stock Returns via Social Media Signals
 - Sentiment Analysis: Dictionary-based Approach
- Analyzing Airline Operations through Twitter Sentiment
 - Sentiment Analysis: LLM-based Approach
 - Azure OpenAI API Calling
- Predicting Short Video Success
 - Machine Learning Pipeline
 - Vector Space Model
 - Topic Modeling
- Social Network Analytics
 - Case Study: Bitcoin Trading
 - Knowledge Graph
- Digital Marketing Analytics
 - Sponsored Ad
 - AIGC / Search Ad Scoring and Generation
- Marketing Campaign Analysis
- Live Streaming E-Commerce
- Business Analytics Project Management
 - Case Study: Facebook Marketing for Movies

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

Required Textbook: None

Reference Books:

1. Dipanjam Sarkar. Text Analytics with Python: A Practitioner's Guide to Natural Language Processing. Apress; 2nd edition (May 22, 2019).
2. Ramesh Sharda, Dursun Delen, Efraim Turban. Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support. Pearson; 11th edition (January 4, 2019).
3. Ryan Mitchell. Web Scraping with Python: Collecting More Data from the Modern Web. O'Reilly Media; Second Edition.
4. Lutz Finger, Soumitra Dutta. Ask, Measure, Learn: Using Social Media Analytics to Understand and Influence Customer Behavior. O'Reilly Media; 1st Edition.
5. Marco Bonzanini. Mastering Social Media Mining with Python. Packt Publishing.
6. Paul Houghton. Data Engineering with Alteryx: Helping data engineers apply DataOps practices with Alteryx. Packt Publishing.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

- ☒ Conducting mid-term survey in additional to SETL around the end of the semester
- ☐ Online response via Moodle site
- ☐ Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

Plagiarism and copying of copyright materials are serious offences and may lead to disciplinary actions. You should read the chapters on "Plagiarism" and "Copyright" in the Undergraduate/Postgraduate Handbook for details. You are strongly advised to read the booklet entitled "What is Plagiarism?" which was distributed to you upon your admission into the University, a copy of which can be found at www.hku.hk/plagiarism. A booklet entitled "Plagiarism and How to Avoid it" is also available from the Main Library.

To avoid intellectual property and copyright infringement, and/or violation of the Personal Data (Privacy) Ordinance, **DO NOT upload** HKU teaching-related materials including but not limited to course materials, marking schemes, examination papers, etc. to websites. If you have done so in the past, you are asked to take steps to take down relevant materials immediately.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Late Penalty

All assignments must be submitted on or before the specified due date and time to the assignment submission destination. A late penalty of 20% deduction per *any fraction* of each 24-hour period of lateness will apply (i.e., if the submission is two days late, 40 out of 100 is automatically deducted for that late submission; zero mark will be given for submissions that are more than 4 days late).